

GrounUp Enterprise Specification

Phase 24 - Core Platform Implementation: Identity, Tenancy, Data, API, Audit & Workflow

Version 1.0.0 | Implementation-ready specification | Technology vendor-neutral

1. Mission

Implement the reusable core platform contracts required before dependent GrounUp business modules are built.

2. Source Alignment

Phase 24 derives its architecture boundaries from Phase 23, which contains 300 foundation requirements and 48 logical architecture components. Phase 24 narrows the build to identity, authorization, tenancy, data access, APIs/events, audit, workflows, configuration and observability.

3. Identity and Authorization

Authentication is provider-neutral. Authorization is default deny and evaluates actor, tenant, resource and applicable attributes/roles/scopes.

4. Multi-Tenant SaaS

Tenant context is mandatory for business data paths. Plans, entitlements, feature flags and white-label configuration are governed configuration rather than duplicated code.

5. Data Foundation

Repositories enforce tenant scope. Schema changes use versioned migrations with validation and rollback/recovery planning.

6. API/Event Foundation

The request pipeline is ordered and governed. Writes use idempotency where retry risk exists. Domain events use a versioned envelope and transactional outbox boundary.

7. Audit and Workflow

Privileged/configuration/approval actions retain actor, tenant, time, correlation, source requirement and evidence metadata. Workflow transitions validate state and authorization.

8. Quality and Security

Cross-tenant negative tests, policy tests, contract tests, security scans, migration tests, observability and traceability are required gates.

9. Implementation Order

Request context -> identity -> tenancy -> authorization/entitlements -> tenant-aware data -> audit/events -> API kernel -> workflow -> config/feature flags -> observability -> E2E acceptance.

Core Modules

ID	Module	Layer	Purpose
P24-MOD-001	Request Context	Platform	Correlation ID, actor, tenant, environment and request metadata
P24-MOD-002	Identity Adapter	Security	Provider-neutral authentication/SSO boundary
P24-MOD-003	Authorization Policy	Security	Default-deny RBAC/ABAC policy evaluation
P24-MOD-004	Tenant Registry	SaaS	Tenant lifecycle, status and provisioning state
P24-MOD-005	Entitlement Service	SaaS	Plans, features, limits and entitlement evaluation
P24-MOD-006	Brand Configuration	SaaS	White-label branding/configuration without business-logic forks
P24-MOD-007	Data Access Boundary	Data	Tenant-aware repository/unit-of-work contracts
P24-MOD-008	Migration Runner	Data	Versioned forward migration, validation and rollback hooks
P24-MOD-009	API Kernel	API	Validation, errors, pagination, idempotency, versioning and middleware
P24-MOD-010	Event Publisher	Events	Versioned domain-event envelope/outbox boundary
P24-MOD-011	Audit Ledger	Governance	Append-oriented governed audit/evidence records
P24-MOD-012	Workflow Kernel	Platform	State transitions, approvals, assignments and escalation hooks
P24-MOD-013	Notification Port	Platform	Provider-neutral notification request boundary
P24-MOD-014	File Metadata Port	Platform	Permission-aware file/object metadata boundary
P24-MOD-015	Feature Flag Service	Platform	Tenant/environment feature evaluation
P24-MOD-016	Configuration Registry	Platform	Versioned platform/tenant configuration
P24-MOD-017	Observability Kernel	Operations	Structured logs, metrics, traces and health
P24-MOD-018	Traceability Hook	Governance	Requirement/test/release evidence metadata